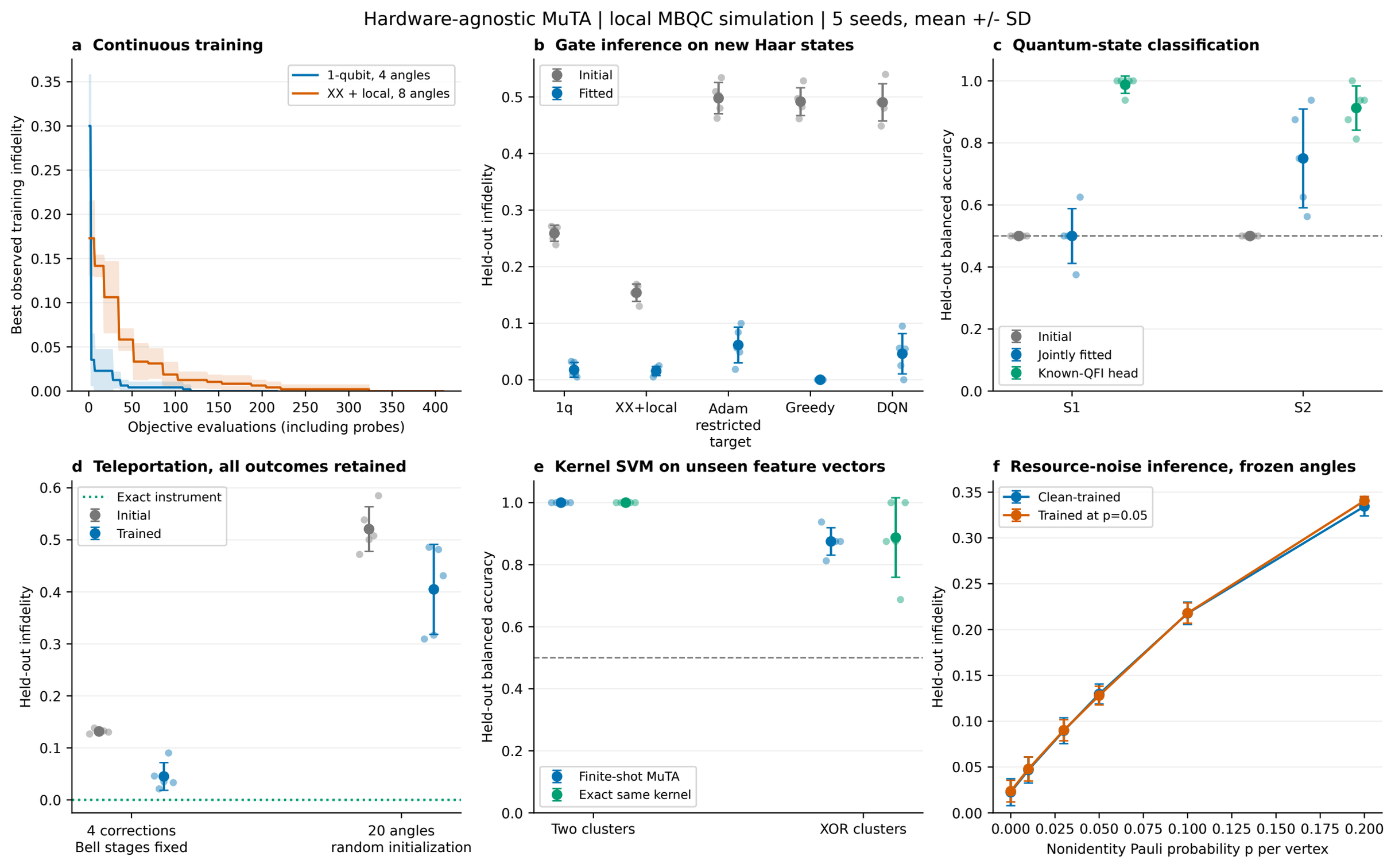

Hardware-agnostic MuTA | local MBQC simulation | 5 seeds, mean +/- SD
a Continuous training
1-qubit, 4 angles
XX + local, 8 angles
Best observed training infidelity
Objective evaluations (including probes)
b Gate inference on new Haar states
Initial
Fitted
Held-out infidelity
1q
XX+local
Adam restricted target
Greedy
DQN
c Quantum-state classification
Held-out balanced accuracy
Initial
Jointly fitted
Known-QFI head
S1
S2
d Teleportation, all outcomes retained
Exact instrument
Initial
Trained
Held-out infidelity
4 corrections Bell stages fixed
20 angles random initialization
e Kernel SVM on unseen feature vectors
Held-out balanced accuracy
Finite-shot MuTA
Exact same kernel
Two clusters
XOR clusters
f Resource-noise inference, frozen angles
Clean-trained
Trained at p=0.05
Held-out infidelity
Nonidentity Pauli probability p per vertex